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Catalytic Transient Leaving Group for Atom-Economic Synthesis of Allenes from 2-Alkynols

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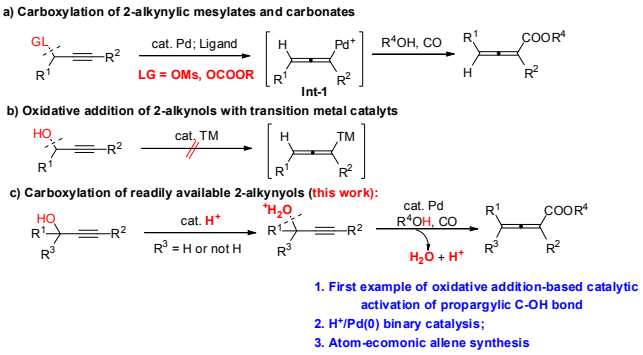
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An atom economic approach from readily available propargylic alcohols to allenes, the first carboxylation of propargylic alcohols, has been established. Under the cooperative binary catalysis of Pd and a phosphoric acid, the reaction afforded multi-substituted allenoates with a broad scope tolerating useful functional groups. Synthetic potentials of obtained products have been demonstrated.

The presence of two cumulated carbon-carbon double bonds is the distinctive structural characteristic of allenes and contributes to their extraordinary reactivities.[1] Due to the broad existence of allene unit in natural products, pharmaceutical molecules, and applications of allenes in advanced molecules/materials, the chemistry of allenes has been catching increasing attention from chemists.[2,3] Especially, much attention has been focused on development of new methodologies for accessing differently substituted allenes recently.[4] In addition to the well established coupling reactions of alkynes with a propargylic leaving group with organometallic reagents,[4] the Pd-catalyzed carboxylative coupling reactions of propargylic mesylates or carbonates with alcohols in the presence of CO afforded 2,3-allenoates via **Int-1**.[5] However, such a coupling strategy utilizing the readily available 2-alkynols has never been reported due to the difficulty encountered in the step of oxidative addition of 2-alkynols with a transition metal catalyst although the S_N2'-type reaction of 2-alkynols with a stoichiometric amount of $TiCl_4$ for the synthesis of 1,2-allenyl chlorides has been reported.[6] We would like to propose a new concept of utilizing a catalytic amount of Brønsted acid to form *a transient leaving group* from the hydroxy group in the readily available propargylic alcohols, which would trigger an oxidative addition with a transition metal catalyst to cleave their propargylic carbon-oxygen bond for subsequent coupling reactions via a similar **Int-1**-type intermediate. The challenge is to identifying the cooperative working Brønsted acid and transition metal catalysts.[7] Herein, we report the first Pd/H^+-cocatalyzed preparation of tetra- or tri-substituted allenoates from 2-alkynols.



Scheme 1. Carboxylation approaches to 2,3-allenoates with CO

Initially, the study was launched using 2-phenyloct-3-yn-2-ol (**1a**) as the starting material, $[(\pi\text{-allyl})PdCl]_2$ as the catalyst and BINAP as the ligand to identify the appropriate acid. When CF_3COOH was added, the expected formations of enone **2a** and enyne **3a** were detected (Table 1, entry 1): enone **2a** was formed through Meyer-Schuster rearrangement and **3a** was formed via dehydration with the help of CF_3COOH (Scheme **2**).[8] A range of acids were then screened, however, the expected product **4a** was not formed (Table 1, entries 2-7). To our delight, when 1,1'-binaphthyl-2,2'-diyl hydrogen phosphate (acid **4**) was used, 22% yield of **4a** was detected together with 7% yield of **3a** (Table 1, entry 8). Using Ph_2POOH led to inferior results-**4a** was formed in only 2% yield (Table 1, entry 9). Surprisingly, the reaction with $(PhO)_2POOH$ afforded 27% yield of allenoate **4a** (Table 1, entry 10). Control experiment was

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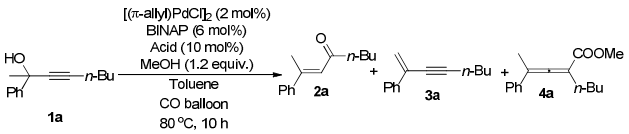
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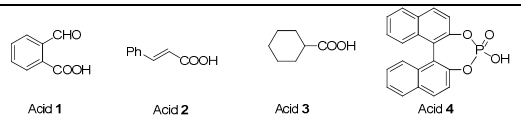
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also conducted: the formation of **2a**, **3a**, and **4a** was not detected in the absence of an acid (Table 1, entry 11).

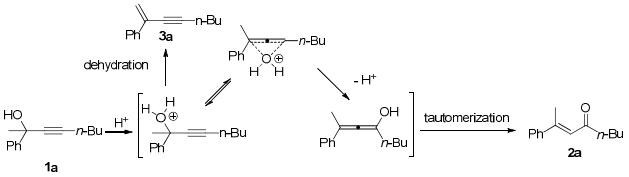
Table 1. Screening of acid with BINAP as the ligand [a]



Entry	Acid	NMR yield (%) [b]			Recovery of 1a (%) [b]
		2a	3a	4a	
1	CF_3COOH	6 [c]	8	-	-
2	AgOTf	27 [c]	-	-	-
3	CH_3SO_2OH	40 [c]	-	-	-
4	BnCOOH	-	-	-	82
5	Acid **1**	-	-	-	80
6	Acid **2**	-	-	-	87
7	Acid **3**	-	-	-	96
8	Acid **4**	-	7	22	-
9	Ph_2POOH	-	1	2	-
10	$(PhO)_2POOH$	-	2	27	5
11	-	-	-	-	71



[a] Reaction conditions: **1a** (0.5 mmol), [(π-allyl)PdCl]$_2$ (2 mol%), BINAP (6 mol%), acid (10 mol%), and MeOH (1.2 equiv.) in toluene (2.0 mL) at 80 °C under 1 atm of CO unless otherwise noted. [b] Determined by ^{1}H NMR analysis using 1,3,5-trimethylbenzene as the internal standard. [c] *E*:*Z* > 20:1.

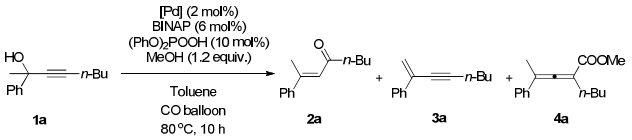


Scheme 2 Mechanism for the formation of enone **2a** and enyne **3a**

Encouraged by these results, different Pd sources were then examined. $PdCl_2$, $PdCl_2(PPh_3)_2$, and [PdCl(cinnamyl)]$_2$ gave 2-21% yields of **4a** (Table 2, entries 1-3). $PdCl_2$(DPPE) and $PdCl_2$(DPPF) were ineffective-nothing but 14-21% yields of **3a** were detected (Table 2, entries 4 and 5). [(π-allyl)PdCl]$_2$ was the most ideal Pd catalyst, obtaining **4a** with 27% yield (Table 2, entry 6). Running the reaction at 50 °C gave better results-a higher yield of **4a** with some recovery of starting material alcohol **1a** was observed (Table 2, entry 7).

Ligands always play an important role in organometallic catalysis. PPh_3 and P(2-furyl)$_3$ were able to accelerate the formation of allenoates, obtaining **4a** with 35-37% yields (Table 3, entries 2 and 3), whereas Zheda-Phos[9] led to nothing but 62% recovery of **1a** (Table 3, entry 4). When Sphos was used, 30% yield of **4a** and 66% recovery of **1a** were detected (Table 3, entry 5). Diphosphine ligands were also investigated: DPPE gave inferior result with 100% of **1a** being recovered (Table 3, entry 6); the desired allenoate **4a** was produced with 20% yield when applying DPPF (Table 3, entry 7); finally, it was found that DPEphos was the most effective ligand, forming **4a** with 64% yield (Table 3, entry 8).

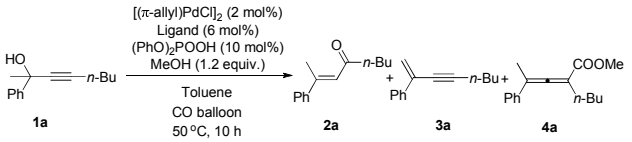
Table 2. The effect of Pd sources [a]



Entry	[Pd]	NMR yield (%) [b]			Recovery of 1a (%) [b]
		2a	3a	4a	
1	$PdCl_2$	-	-	2	-
2	$PdCl_2(PPh_3)_2$	-	-	2	-
3	[PdCl(cinnamyl)]$_2$	-	5	21	-
4	$PdCl_2$(DPPE)	-	21	-	-
5	$PdCl_2$(DPPF)	-	14	-	-
6	[(π-allyl)PdCl]$_2$	-	-	27	5
7 [c]	[(π-allyl)PdCl]$_2$	-	1	42	13

[a] Reaction conditions: **1a** (0.5 mmol), [Pd] (2 mol%), BINAP (6 mol%), $(PhO)_2POOH$ (10 mol%), and MeOH (1.2 equiv.) in toluene (2.0 mL) at 80 °C under 1 atm of CO unless otherwise noted. [b] Determined by ^{1}H NMR analysis using 1,3,5-trimethylbenzene as the internal standard. [c] The reaction was carried out at 50 °C.

Table 3. The effect of ligand [a]



Entry	Ligand	NMR yield (%) [b]			Recovery of **1a** (%) [b]
		2a	3a	4a	
1	BINAP	-	1	42	13
2	PPh_3	-	-	35	47
3	P(2-furyl)$_3$	50 [c]	11	37	-
4	Zheda-phos	-	-	-	62
5	Sphos	-	-	30	66
6	DPPE	-	-	-	100
7	DPPF	-	6	20	44
8	DPEphos	-	-	64	10
9 [d]	DPEphos	-	-	86	-

[a] Reaction conditions: **1a** (0.5 mmol), [(π-allyl)PdCl]$_2$ (2 mol%), Ligand (6 mol%), $(PhO)_2POOH$ (10 mol%), and MeOH (1.2 equiv.) in toluene (2.0 mL) at 50 °C under 1 atm of CO unless otherwise noted. [b] Determined by ^{1}H NMR analysis using 1,3,5-trimethylbenzene as the internal standard. [c] *E*:*Z* > 20:1. [d] 5 mol% $(PhO)_2POOH$ and 8.0 equiv. MeOH were used instead.

After further optimization, it is apparent that when 2 mol% [(π-allyl)PdCl]$_2$, 6 mol% DPEphos, 5 mol% $(PhO)_2POOH$, and 8.0 equiv. MeOH were applied in toluene at 50 °C under 1 atm of CO (Table 3, entry 9, defined as the standard conditions), 2,3-allenoates can be obtained with the highest yield from the 2-alkynylic alcohols (for details, see Tables S2 and S3 in Supporting Information). The substrate scope was then evaluated (Table 4). When R^1 is an alkyl group, R^2 may be alkyl and phenyl or substituted phenyl (Table 4, **4a**-**4n**). The steric effect is obvious since 2-methylphenyl led to a much lower

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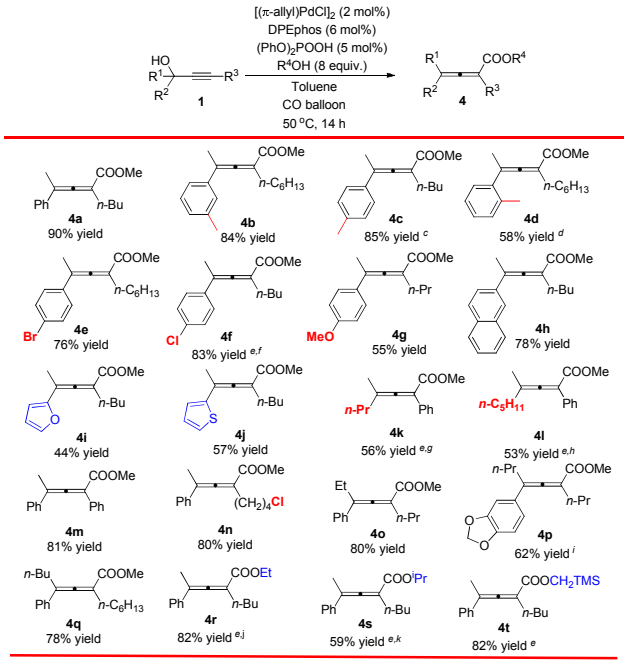
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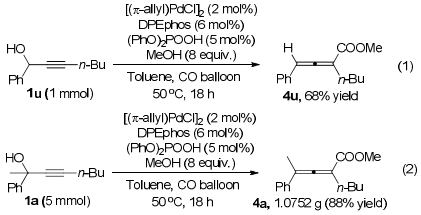
yield (compare **4b**, **4c**, and **4d**); it is interesting to observe that the current reaction conditions are selecttive enough to differentiate the reactivity of C-Br bond and the propargylic C-O bond in **1e**: the C-Br bond remained untouched for further elaboration. In addition, functional groups such as chloro, and methoxy are well tolerated under the optimized reaction conditions (Table 4, **4f and 4g**); heteroaryl groups such as furyl and thienyl made no difference, furnishing the target products with moderate yields (Table 4, **4i** and **4j**). R^3 may also be an alkyl or aryl group. Meanwhile, reactions of ethanol, isopropanol, and trimethylsilylmethanol have also been studied, providing allenoates possessing different kinds of R^4 group with good yields (Table 4, **4r**-**4t**).

Table 4. Substrate scope for the synthesis of multi-substituted allenoates from propargylic alcohols [a]

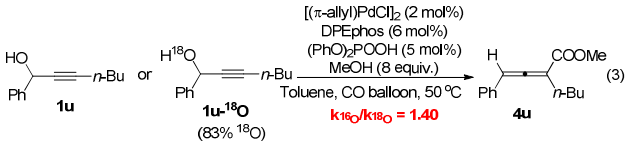


[a] Reaction conditions: **1** (1.0 mmol), [(π-allyl)PdCl]$_2$ (2 mol%), DPEphos (6 mol%), $(PhO)_2POOH$ (5 mol%) and MeOH (8 equiv.) in toluene (5.0 mL) at 50 °C under 1 atm of CO unless otherwise noted. [b] isolated yields. [c] the reaction was carried out for 10 h. [d] 36% recovery of **1d** was detected. [e] the reaction was carried out with 16 equiv. of R^4OH. [f] the reaction was carried out for 16 h, 8% recovery of **1f** was detected. [g] 32% recovery of **1k** was detected. [h] 44% recovery of **1l** was detected. [i] the reaction was carried out with 3 mol% [(π-allyl)PdCl]$_2$ and 9 mol% DPEphos. [j] 11% recovery of **1a** was detected. [k] 20% recovery of **1a** was detected.

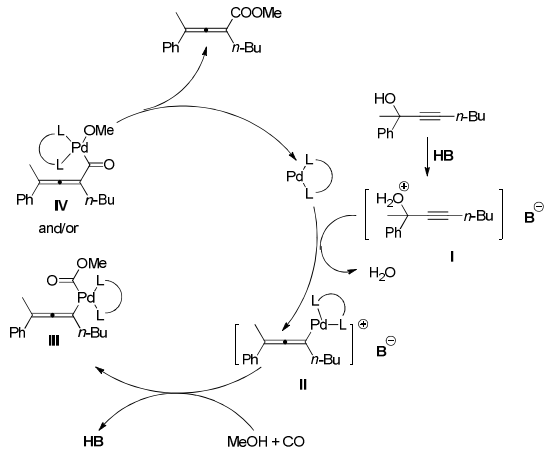
Apart from these, secondary propargylic alcohol may also be utilized to give trisubstituted allenoates **4u** with 68% yield (eq. 1). The reaction may be easily conducted on 5 mmol scale of **1a** (eq. 2).



In order to unveil the mechanism of this carboxylation reaction, the reactions of **1u** and **1u-^{18}O** at 50 °C were conducted and analyzed via ^{1}H-NMR analysis (For details, see Supporting Information). A linear relationship was observed for ln ([**1u**]) vs. reaction time, indicating a first-order dependence of reaction rate with respect to substrate **1u**.[10] In addition, no **^{18}O** was found in the product **4u**. The kinetic isotopic effect was determined to be 1.40, indicating that the cleavage of C-O bond is not the rate-determined step (eq. 3).



Based on these results, a plausible mechanism is proposed (Scheme 3): the reaction of propargylic alcohol with Brønsted acid (HB) would yield the oxonium ion I, which would easily undergo oxidative addition reaction with Pd(0) to produce the allenylpalladium intermediate II. Its subsequent reaction with CO and MeOH afforded intermediates III and/or IV while regenerating HB. Although intermediate III may be more viable,[11] complete exclusion of intermediate IV is not possible. Subsequent reductive elimination would afford 2,3-allenoate products and regenerate active Pd(0) catalyst to fulfill the catalytic cycle. We reasoned that the acidity of $(PhO)_2POOH$ matches the propargylic alcohols and the large bite angle of DPEphos (104°) (DPPE (86°) and DPPF (99°))[12] accelerated the reductive elimination of intermediate III and/or IV.

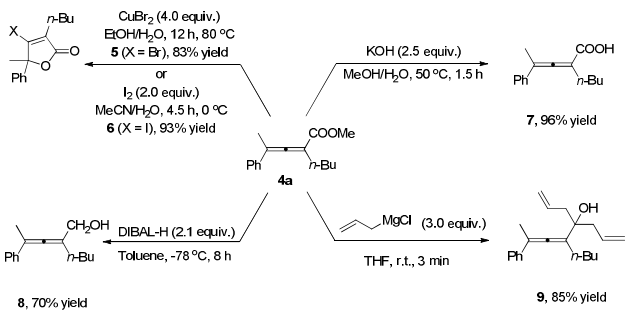


Scheme 3. A plausible mechanism

As a class of useful chemicals in organic synthesis, 2,3-allenoates can be transferred into different versatile products (Scheme 4): treatment with $CuBr_2$ or I_2 afforded 4-bromo-2(*5H*)-furanone (**5**)[13] or 4-iodo-2(*5H*)-furanone (**6**)[14] with good to excellent yields; hydrolysis with KOH at 50 °C for 1.5 hours afforded tetra-substituted allenic acid **7** in a nearly quantitative yield;[15] reduction with DIBAL-H at -78 °C could yield 2,3-allenol **8** with 70% yield; 1,2-addition reaction with allyl magnesium chloride furnished tertiary *α*-allenol **9** with 85% yield.[16]

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Scheme 4. Synthetic applications of tetra-substituted allenotae **4a**

In conclusion, a robust and atom-economic straightforward protocol utilizing Brønsted acid and palladium as the catalysts for the preparation of multi-substituted 2,3-allenoates from the readily available propargylic alcohols with moderate to good yields directly was developed. The synthetic potentials of these allenoates have also been demonstrated forming other useful building blocks. Mechanistic study shows that the cleavage of C-O bond with the k_{16O}/k_{18O} of 1.40. This protocol would open a new area for the synthesis of allenes directly from propargylic alocols via the **Int-1**-based coupling strategy. Further studies in this area including the role of each catalyst and enantioselective carboxylation are being conducted in our laboratory.

ACKNOWLEDGMENTS

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Conflicts of interest

There are no conflicts to declare.

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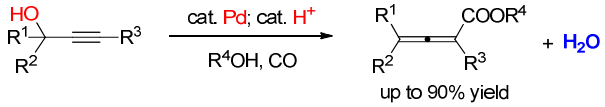
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